

**Problem Set 2**  
Winter 2020

**Issued:** Saturday, January 18

**Due:** Saturday, February 1

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**Notation:** The symbols  $\{\xrightarrow{d}, \xrightarrow{p}, \xrightarrow{m.s.}\}$  denote convergence in distribution, probability and mean-square respectively.

**Probability background:** Keener Chapter 1 and Sections 6.1, 6.2, 8.1, and 8.2.

**Reading:** Chapter 3

Full credit: 100 points.

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**Problem 2.1** (25 points)

Consider the *normal means model*  $x = \theta + z \in \mathbb{R}^n$  where  $z \sim N(0, I_n)$ . That is,

$$x_i = \theta_i + z_i, \quad z_i \stackrel{\text{iid}}{\sim} N(0, 1), \quad i = 1, \dots, n.$$

We treat  $\theta \in \mathbb{R}^n$  as an unknown parameter and we want to estimate it based on the observation  $x$ .

Let  $\hat{\theta} = \hat{\theta}(x)$  be a general estimator and consider its quadratic risk  $R(\theta, \hat{\theta}) = \mathbb{E}_\theta[\|\hat{\theta} - \theta\|^2]$  where  $\|y\| = \sqrt{\sum_{i=1}^n y_i^2}$  is the usual  $\ell_2$  Euclidean norm of vector  $y \in \mathbb{R}^n$ .

(a) Show that  $\text{cov}_\theta(\hat{\theta}_i, x_i) = \mathbb{E}_\theta[\partial \hat{\theta}_i / \partial x_i]$ . (*Hint: Integration by parts.*)

(b) Show that

$$R(\theta, \hat{\theta}) = \mathbb{E}_\theta[\|x - \hat{\theta}\|^2] - n + 2 \sum_{i=1}^n \mathbb{E}_\theta \left[ \frac{\partial \hat{\theta}_i}{\partial x_i} \right].$$

(c) Obtain  $\hat{\theta}^{(MLE)}$ , the maximum likelihood estimator (MLE) of  $\theta$ , and compute its risk.

(d) Consider the James-Stein estimator (JSE),

$$\hat{\theta}^{(JSE)} := \left( 1 - \frac{n-2}{\|x\|^2} \right) x,$$

and show that

$$R(\theta, \hat{\theta}^{(JSE)}) = R(\theta, \hat{\theta}^{(MLE)}) - \mathbb{E}_\theta \left\{ \frac{(n-2)^2}{\|x\|^2} \right\}.$$

(e) Argue that for  $n \geq 3$ , one of the JSE and MLE renders the other inadmissible for quadratic loss (and specify which one).

**Problem 2.2** (5 points)

Consider an i.i.d. model  $X_1, \dots, X_n \stackrel{iid}{\sim} p(x; \theta)$ .

Show that the order statistic  $T = (X_{(1)}, \dots, X_{(n)})$  is sufficient.

Recall that  $T$  has the same values as  $(X_1, \dots, X_n)$  but ordered as  $X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$ .

**Problem 2.3** (15 points)

Consider the two parameter exponential distribution  $E(\mu, \lambda)$ , with density

$$\lambda e^{-\lambda(x-\mu)} 1\{x \geq \mu\}.$$

Let  $X_1, \dots, X_n \stackrel{iid}{\sim} E(\mu, \lambda)$ . Show that  $T = (X_{(1)}, \sum_i [X_i - X_{(1)}])$  is minimal sufficient, for parameter  $\theta = (\mu, \lambda)$ .

**Problem 2.4** (20 points)

Consider the characterization of conditional expectation as  $L^2$  projection, that is,  $\hat{g}(X) := \mathbb{E}[Y|X]$  is the (essentially) unique function of  $X$  satisfying

$$\mathbb{E}[(Y - \hat{g}(X))g(X)] = 0$$

for any (meas.) function  $g(\cdot)$  such that  $g(X) \in L^2$ . Using this characterization prove the following 4 items:

(a)  $\mathbb{E}[Yh(X) | X] = h(X) \mathbb{E}[Y|X]$ .

Let  $X = (X_1, X_2, X_3, \dots)$ , and for any subset  $S \subset \mathbb{N}$ , let  $X_S = (X_j, j \in S)$ . For example,  $X_{\{1,3,5\}} = (X_1, X_3, X_5)$ .

(b)  $\mathbb{E}[\mathbb{E}[Y|X_S] | X_T] = \mathbb{E}[Y|X_S]$  whenever  $S \subset T$ .

(c)  $\mathbb{E}[\mathbb{E}[Y|X_S] | X_T] = \mathbb{E}[Y|X_T]$  whenever  $T \subset S$ .

Recalling that the  $L^2$  norm is given by  $\|Y\|_{L^2} := \sqrt{\mathbb{E}[Y^2]}$ ,

(d)  $\|\mathbb{E}[Y|X]\|_{L^2} \leq \|Y\|_{L^2}$ . (Conditional expectation is a *nonexpansive* operator.)

Assume that all the RVs are in  $L^2$  (i.e., have finite second moments).

**Problem 2.5** (20 points)

Suppose that  $(X_1, X_2, \dots, X_n)$  is multivariate normal with unknown mean  $(\mu, \dots, \mu) \in \mathbb{R}^n$  and covariance matrix  $\sigma^2 I$ . First, suppose that  $\sigma^2 > 0$  is known. Show that  $T = \sum_{i=1}^n X_i$  is sufficient for the parameterization  $\theta = \mu$  in two ways:

(a) using the factorization criterion, and

(b) by direct use of the definition of sufficiency.

Now suppose that  $\sigma^2 > 0$  is not known, and the model is parameterized by  $\theta = (\mu, \sigma^2)$ . Find and establish sufficiency of a non-trivial set of sufficient statistics (i.e., not the data itself).

**Problem 2.6** (20 points)

Consider the family of variates  $(X_1, X_2, \dots, X_n)$  with  $X_i \sim \text{Uni}[\theta - \frac{1}{2}, \theta + \frac{1}{2}]$ .

- (a) Show that  $T = (X_{(1)}, X_{(n)})$  is sufficient, where  $X_{(1)} = \min X_i$  and  $X_{(n)} = \max X_i$ .
- (b) Is  $T$  minimal sufficient?
- (c) Suppose that we wish to estimate  $\theta$  under the quadratic loss  $L(\theta, a) = (\theta - a)^2$ . The sample mean  $\bar{X}$  is one reasonable estimate of  $\theta$ , but turns out to be inadmissible. Show that the estimator

$$\delta(X_1, \dots, X_n) := \mathbb{E}[\bar{X} \mid X_{(1)}, X_{(n)}]$$

has a strictly better MSE than the MSE of  $\bar{X}$ , and moreover that  $\delta(X_1, \dots, X_n) = \frac{1}{2}[X_{(1)} + X_{(n)}]$ .

(Hint: The conditional distribution of  $X_i$  given  $(X_{(1)}, X_{(n)})$  is not uniform.)

**Problem 2.7** (15 points)

Consider an exponentially distributed variate with density

$$f_X(t) := \begin{cases} \lambda \exp(-\lambda t) & \text{for } t > 0 \\ 0 & \text{otherwise.} \end{cases}$$

The parameter  $\lambda > 0$  is known as the failure rate.

- (a) Suppose  $X$  has an exponential distribution with failure rate  $\lambda$ . Find the distribution of  $Y = \lambda X$ ;
- (b) Let  $X_1, \dots, X_n$  be a sample from an exponential distribution, and let  $\bar{X} = \frac{X_1 + \dots + X_n}{n}$ . Show that  $\bar{X}$  and  $\frac{(X_1^2 + \dots + X_n^2)}{\bar{X}^2}$  are independent.
- (c) Let  $X_{(1)} \leq \dots \leq X_{(n)}$  be the order statistics and  $\bar{X} = \frac{X_1 + \dots + X_n}{n}$  the sample average. Show that  $\bar{X}$  and  $\frac{X_{(1)}}{X_{(n)}}$  are independent.